

ENGR 1451/2451: Exploratory Data Science

Correlation, Confounding, and Causation

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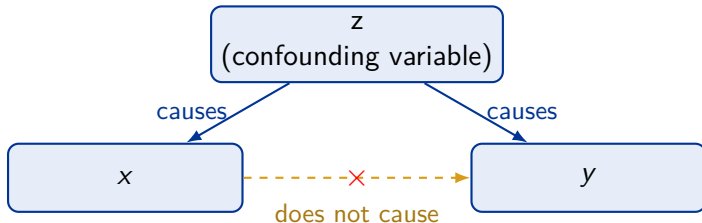
Today: Association and Causation

Correlation and regression describe how variables vary together. This lecture examines when a relationship supports a **causal claim**.

- Distinguish **association** from **cause and effect**.
- Compare **controlled experiments** with **observational studies**.
- Identify **confounding** and explain how randomization and careful comparisons help.
- Use **Simpson's paradox** to understand why overall and subgroup comparisons can differ.

Correlation does not imply causation

- Correlation measures association.
- However, association does not necessarily show causation.
- Both variables may be simultaneously influenced by some third variable.
- That third variable is known as a **confounding variable**



Ice Cream Sales and Drownings

Observed correlation

In many places, monthly **ice cream sales** and **deaths by drowning** tend to rise and fall together.

A tempting claim

More ice cream consumption causes more drownings.

Consider the explanations

What are at least **two** possible explanations for this correlation?

Controlled Experiments

- Controlled experiments compare an **experimental group** with a **control (baseline) group**.
- Ideally, the groups are similar in all important respects except for the **factor being changed**, or **predictor variable (x)**.
- If the groups are otherwise similar, a difference in the **response variable (y)** can reasonably be attributed to the change in x .
- If the groups also differ in other important ways, these factors may be **confounded** with the predictor variable.
- Confounding makes it difficult to determine whether the observed change in y was actually caused by x .

Randomized Experiments in Engineering

- **Randomization** is often used to reduce confounding and systematic bias.
- Experimental units—such as parts, samples, devices, or test runs—are randomly assigned to the **experimental condition** or **control condition**.
- Randomization helps distribute both known and unknown sources of variation across the groups.
- Good experimental design combines **control**, **randomization**, and **consistent measurement procedures** to isolate the effect of x on y .

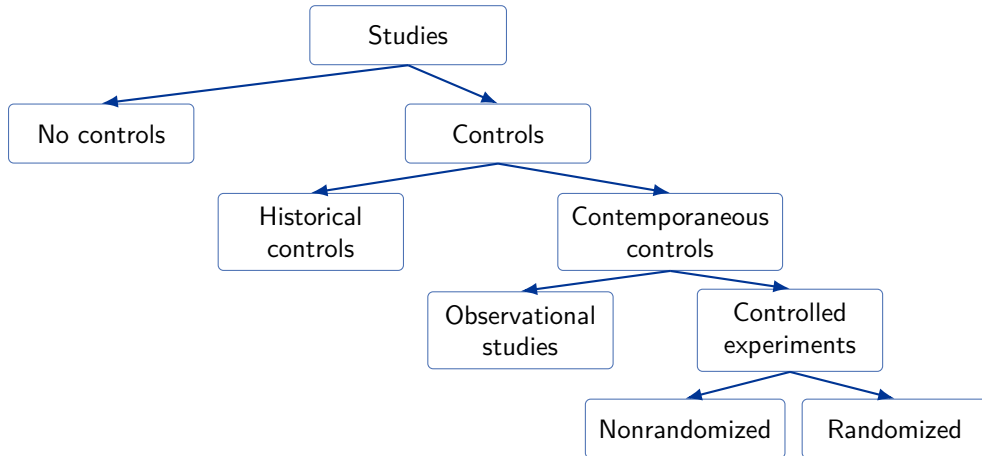
Medical Studies

- Subjects are randomly assigned to the **treatment group** or **control group**. Randomization helps make the groups similar with respect to other factors.
- When possible, the control group receives a **placebo**: an inactive treatment that resembles the actual treatment.
- A placebo helps separate the effect of the treatment from the effect of *believing* that a treatment was received.
- In a **double-blind** experiment, neither the subjects nor the researchers evaluating them know who received the treatment, helping reduce bias.

Observational Studies

- In an **observational study**, investigators do not assign subjects to treatment or control groups. Instead, they observe groups that differ naturally in the treatment, exposure, or condition of interest.
- Observational studies can identify **associations**: changes in one variable are related to changes in another.
- An association may suggest a possible causal relationship, but **association alone does not establish causation**.
- A major challenge is **confounding**. A third variable may be related to both the treatment or exposure and the response, making it difficult to determine what actually caused the observed difference.
- Confounding can sometimes be reduced by comparing groups that are similar with respect to important confounding variables.

Design of Experiments



Salk Vaccine: Two Designs

National Foundation for Infantile Paralysis (NFIP) design

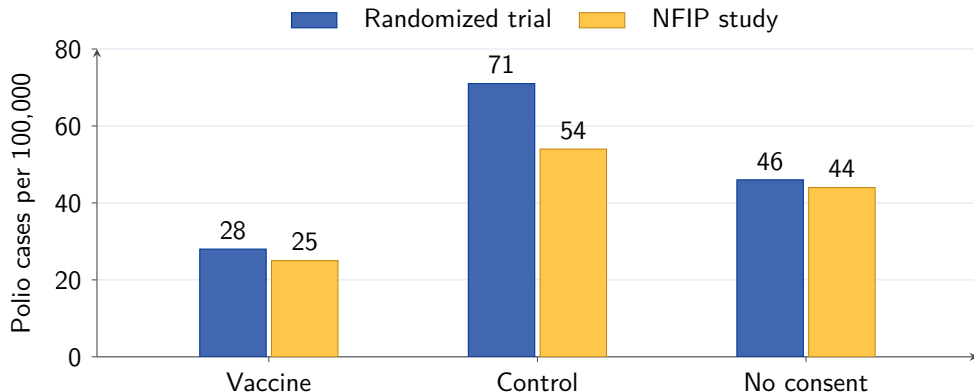
- Vaccinate grade 2 children whose parents consented.
- Use grades 1 and 3 as controls.
- No placebo or blinding.

Randomized controlled double-blind design

- Start with children whose parents consented.
- Randomly assign vaccine or placebo.
- Keep children and diagnosticians blind to assignment.

Which design was better?

Salk Vaccine Trial Results



Ultrasound and Birthweight

Observed association

In an observational study, babies whose mothers received ultrasound examinations had **lower birthweight**, on average.

The association remained even after adjusting for several measured variables.

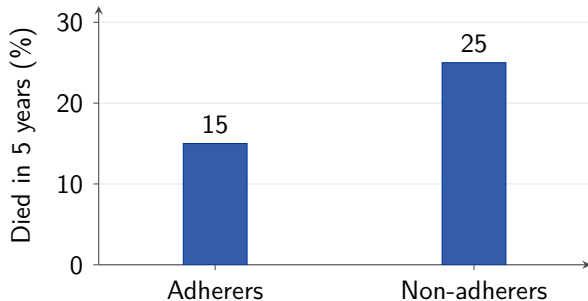
Does this mean ultrasound causes lower birthweight?

Why might patients who received ultrasound differ systematically from those who did not?

Historical Controls

- A **historical control** is a control group taken from past data rather than from subjects observed at the same time as the treatment group.
- **Medical example:** A hospital evaluates a new surgical procedure by comparing current patients with patients who received the previous procedure several years ago.
- **Engineering example:** A manufacturer compares the failure rate of a redesigned component with historical failure data from the previous design.
- **Problem:** Other conditions may have changed over time, so differences between the groups may be **confounded with time**.

Clofibrate: Does Taking the Drug Help?

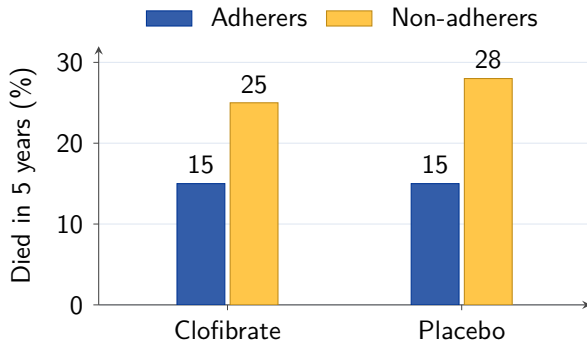


Among patients assigned clofibrate

Patients who took the drug regularly had substantially lower mortality than those who did not.

Does this show that taking clofibrate reduced mortality?

Clofibrate: Why Do We Need a Control Group?



What does the placebo group show?

Placebo adherers also had much lower mortality than placebo non-adherers.

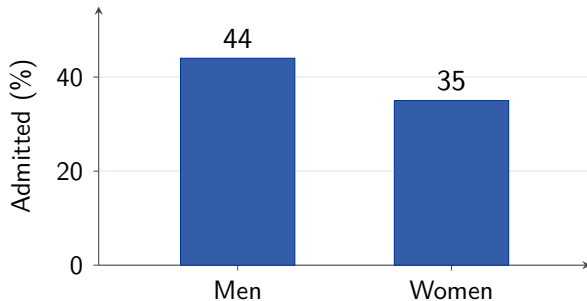
The same pattern appears even when the “treatment” is a placebo.

So adherence itself is associated with mortality.

Clofibrate table and interpretation from Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), Ch. 2, pp. 13–14, Table 1..

Berkeley Graduate Admissions

What could differ between male and female applicants besides sex?



Question

Does this prove that the University discriminated against women?

Berkeley Admissions: Control for Department

Major	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

Look across rows. Do you still see a large disadvantage for women within departments?

What changed after controlling for department?

Within the largest departments, male and female admission rates were much closer. Some departments favored women; others favored men.

Simpson's Paradox: How the Reversal Happened

Different application patterns

- Departments A and B were much easier to enter.
- Over half of male applicants applied to A or B.
- Over 90% of female applicants applied to the harder departments C–F.

Adjust for department

Imagine that men and women applied to the six departments in **exactly the same proportions**.

Use the same department mix to average the admission rates for each group:

Men: 39% Women: 43%

Why did the comparison reverse?

The overall rates mixed together departments with very different admission rates. Men applied more often to easier departments, while women applied more often to harder departments.

**After controlling for department,
the apparent overall advantage for men disappears.**

Summary

- **Association alone does not establish causation.** A confounder may explain an observed relationship.
- **Randomized experiments provide stronger causal evidence.** Random assignment helps balance confounders. Blinding and placebos can reduce bias.
- **Observational studies require careful comparisons.** Groups form through choice or circumstance. Adjustment can reduce confounding, but important differences may remain.
- **Overall and subgroup comparisons can differ.** Simpson's paradox shows that an association can reverse when groups are combined.